

Frontend

PyTorch Model

KAN / MLP import
State dict extraction
B-spline coeff unpack

SI units recognition

Python

IR Graph

6 Operation Types

MatMul
StandardAct
Add

BsplineLU
Softmax
Argmax

Typed attrs + Topo sort
Cycle detection

JSON DAG

Backend

S7-1200 Target

50 KB work memory
DB + FB SCL split
15-pt LUT, FOR-loop

S7-1500 Target

1.5 MB budget
50-pt LUT, unrolled

IEC 61131-3 SCL

Validation

TIA Portal V21

MCP Openness API
Compile & Diagnose

0 errors · 0 warnings

Cross-Validation

Python vs SCL LUT
100% class agreement

Optimizer (6 Passes)

Adaptive B-spline LUT

FuseMatMulAdd

Dead-Node Elim.

Code-level transforms, 0 accuracy

Static Analyzer

Memory budget (KB)

FLOPs + Instructions

WCET bound (Z3 SMT)

DA Safety Factor

TIA V21 measured values

Verification (3-Tier)

Tier 1: Template Proofs (4/6 ops)

Tier 2: Per-Func Z3 (512/512 UNSAT)

Tier 3: Cor

Legend

- Data flow (pipeline)
- > Cross-cutting concern
- Compiler frontend/optimizer
- IR / static analysis
- Backend / code generation
- Validation / deployment
- Verification / certification